ZentroMart VoiceOps

Capstone Project Report

Synthetic customer support with retrieval, guarded tools, and Gnani voice adapters

Prepared	Dataset reference date	Selected LLM
22 October 2026	20 October 2026	Gemini Flash-Lite

Executive summary

This capstone builds a customer-support assistant in four stages: a prompt-only Gemini baseline, retrieval over the supplied synthetic company knowledge base, a multi-turn agent with verified order lookup and guarded ticket creation, and Gnani speech-to-text/text-to-speech adapters. The RAG stage completed its full 52-question practice set with all 47 answerable questions finding their expected evidence at top five and a reviewed answer pass rate of 52/52. The baseline passed 8/52 literal required-fact checks. The first complete multi-turn run passed 82/96 turns before several fixes; targeted replays later confirmed corrections in selected cases. A complete post-fix replay was blocked by Gemini HTTP 429 free-tier rate limiting. The synthetic single-turn voice flow and one-file batch test passed using Gnani official SDK. STT transcribed generated speech, Gemini returned a support reply, and TTS generated a valid WAV reply. The batch runner can retain state across ordered turns. A user-recorded sample, multi-file voice verification, and demo video remain to validate.

Problem and goals

Customers ask general policy questions and questions about specific orders. The assistant must answer from company evidence, verify identity before disclosing order details, avoid leaking personal data, and create a human handover only for qualifying cases. The final voice flow must transcribe audio, run the support flow, synthesize a spoken reply, and save evidence and tool-call records for each recording.

Implementation

Stage 1 - Prompt-only baseline

Gemini Flash-Lite answers without retrieval. This creates a simple baseline and demonstrates the limits of general model knowledge. Provider credentials are read from a local .env file and are excluded from the submission.

Stage 2 - Retrieval-augmented generation

A local BM25-style retriever indexes policy sections and approved catalog/pincode tables. Search expansions cover common English, Hindi, and Hinglish variants. Retrieved source IDs are included in the answer flow. Business-day refund dates use the policy duration and holiday calendar; return deadlines use calendar-day arithmetic.

Stage 3 - Multi-turn tools

The agent stores conversation state in memory and uses the dataset local mock tools for order lookup and ticket creation. Specific order details require an order ID plus the phone last four digits or delivery pincode. Verification proof and direct contact fields are removed from model context and ticket summaries. Ticket creation is gated on an explicit human request or policy escalation. These mock tools are educational, not production services.

Stage 4 - Voice

Gnani STT and TTS use the official gnani-vachana Python SDK. The batch runner accepts mono 16 kHz, 16-bit PCM WAV and saves transcript, assistant text, synthesized WAV, tool-call record, and retrieved evidence IDs. The single-turn and one-file batch paths ran end to end with synthetic voice through the official SDK: STT transcribed the input, Gemini generated a support response, and TTS saved a valid reply WAV and batch record. The batch runner also supports one session across sorted audio turns with --conversation. A user recording, multi-file voice order-verification dialogue, and the short demo video remain to be completed.

Evaluation results

Stage	Evaluation	Result	Interpretation
Prompt-only	52 practice questions	8/52	Literal fact check; no knowledge base; 49/52 escalated conservatively.
Retrieval	47 answerable; top-k 5	47/47	All expected evidence found for the answerable subset.
RAG	52 practice questions	52/52	Reviewed for multilingual paraphrases/date math; 47/47 expected sources found.
Multi-turn	96 turns; initial full run	82/96	Pre-fix snapshot. Targeted corrections tested; post-fix full rerun blocked by HTTP 429.
Gnani voice	One synthetic input; single + batch	Passed	Official SDK STT, Gemini reply, valid TTS WAV, and saved batch outputs. Real recording and demo video remain pending.

Interpretation limits. The practice set is synthetic and small. RAG includes manual review of multilingual paraphrases, so 52/52 is not an independent human study. The 82/96 multi-turn score is from the initial full run before the final fixes; it is not a post-fix result. Targeted live replays verified selected cases, including standard-delivery compensation refusals and shipped-order cancellation. Gemini free-tier rate limiting prevented a complete post-fix replay.

Run instructions

Use Python 3.10 or newer, install requirements.txt, extract the supplied dataset to work/dataset/Capstone_Project_Dataset_FTTT_2026, copy .env.example to .env, and set LLM_API_KEY locally. Voice features also require GNANI_API_KEY. Never commit .env.

```
python run_voice_batch.py path/to/audio_folder --language en-IN --voice-language auto
```

Voice inputs must be mono 16 kHz, 16-bit PCM WAV. The supplied dataset contains no audio recordings. Full baseline, retrieval, RAG, and tool-agent commands are in the project README.

Submission status and next steps

The source package is prepared without the dataset, virtual environment, generated ticket file, or secrets. A public GitHub repository, deployed demo, and the required demo video (maximum two minutes) have not been created. To finish submission, record a real voice interaction and a maximum two-minute demo showing a document-grounded question with a follow-up plus one handover or safe refusal. Test multiple self-recorded WAV turns with run_voice_batch.py --conversation and inspect transcripts, reply audio, and evidence IDs. Rerun the full post-fix text evaluation after Gemini quota is available.

Sources: supplied VoiceOps 2026 Capstone Brief and synthetic FTTT dataset (README, policy files, practice data, tables, and mock tools); Gnani official Python SDK (gnani-vachana) and API documentation at docs.gnani.ai/api; Google AI Studio Gemini API.